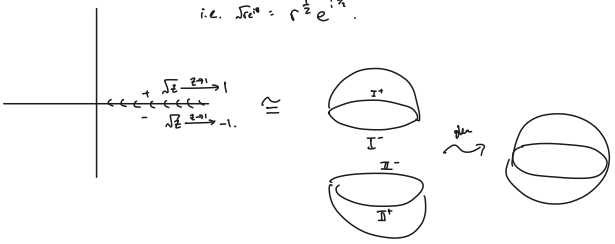
Want define $\omega = \sqrt{z}$ on a max. domain of hol.

$\sqrt{z} := e^{\frac{1}{2}\log z}$, when $\log : \mathbb{C} \setminus \mathbb{R}_{\geq 0}$

i.e. $\sqrt{re^{i\theta}} = r^{\frac{1}{2}} e^{i\theta/2}$.



$$|(-z)^{s-1}| = e^{(\sigma-1)\log|z| - \tau \arg(-z)}$$
$$= \frac{(\sqrt{z}\,\epsilon)^{\sigma-1}}{e^{\tau \arg(-z)}}.$$

$$|\arg(-z)| \leq \pi$$

z :

$-z$: $\rightsquigarrow \arg(-z) \leq 2\pi$

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1. $\omega_1, \omega_2 \in \mathbb{C}$ lin. ind./$\mathbb{R}$, $\tau = \frac{\omega_2}{\omega_1}$. $\operatorname{Im}\tau > 0$. $X_\tau = \mathbb{C}/\mathbb{Z}+\mathbb{Z}\tau$.

$\sigma = \frac{a\tau+b}{c\tau+d}$, $a,b,c,d \in \mathbb{Z}$, $ad-bc=1$. Show $\operatorname{Im}(\sigma) > 0$ and X_σ biho. X_τ.

Proof. $\operatorname{Im}(\sigma) = \frac{\sigma - \bar{\sigma}}{2i} = \frac{\frac{a\tau+b}{c\tau+d} - \frac{a\bar{\tau}+b}{c\bar{\tau}+d}}{2i}$

$$= \frac{(a\tau+b)(c\bar{\tau}+d) - (a\bar{\tau}+b)(c\tau+d)}{(c\tau+d)(c\bar{\tau}+d)\,2i}$$

$$= \frac{ac|\tau|^2 + ad\tau + bc\bar{\tau} + bd - ac|\tau|^2 - ad\bar{\tau} - bc\tau - bd}{2i\,(c^2|\tau|^2 + cd\tau + cd\bar{\tau} + d^2)}$$

$$= \frac{(ad-bc)(\tau - \bar{\tau})}{2i\,|c\tau+d|^2}$$

$$= \frac{\operatorname{Im}(\tau)}{|c\tau+d|^2} > 0$$

ϕ. $X_\tau \longrightarrow X_\sigma$

$[z] \longmapsto \left[\frac{az+b}{cz+d}\right]$

note $n_1 + n_2\tau \longmapsto \frac{an_1 + an_2\tau + b}{cn_1 + cn_2\tau + d}$

$\equiv 0 \mod \mathbb{Z} + \mathbb{Z}\sigma$.

want $cz+d \neq 0$

$z \neq -\frac{d}{c} \in \mathbb{R} \longrightarrow$ can find representative.

So ϕ hol. inj: left inverse

$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \rightsquigarrow \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ $\quad \frac{az+b}{cz+d} \longmapsto \frac{d\left(\frac{az+b}{cz+d}\right) - b}{-c\left(\frac{az+b}{cz+d}\right) + a}$

$$= \frac{adz + db - bcz - bd}{-acz - bc + acz + ad}$$

$$= z.$$

Surj:

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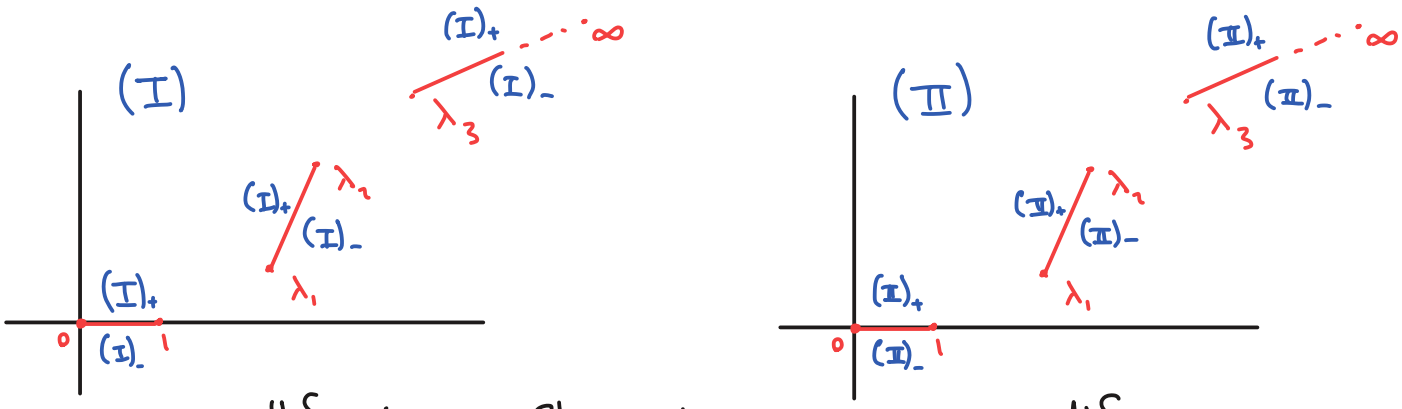
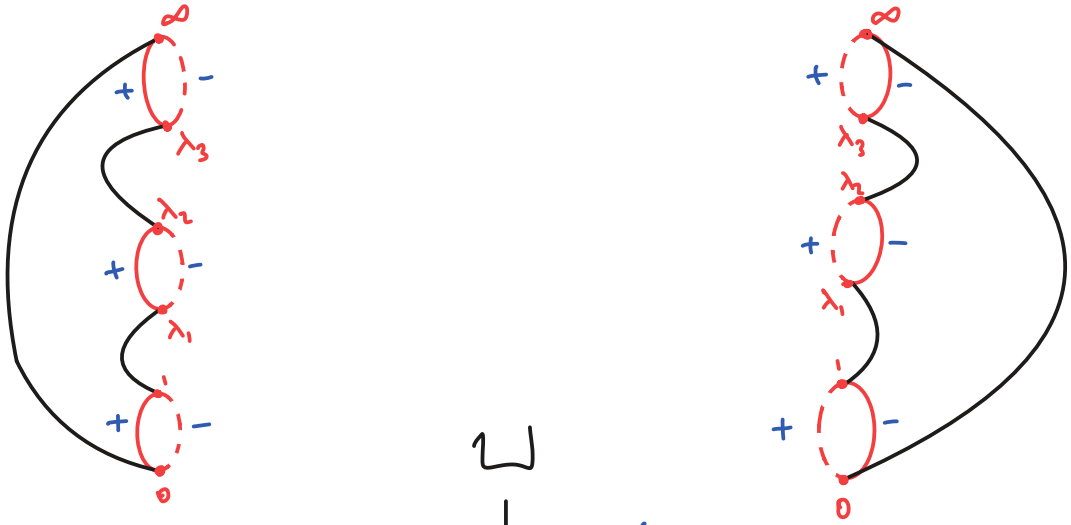
$0, 1, \lambda_1, \lambda_2, \lambda_3 \in \mathbb{C}$ pairwise distinct.



$\cong S \xleftarrow{\text{stereographic projection}} \cong S$

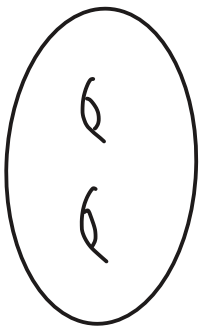
$\cong S \qquad \cong S$



$\sqcup$

glue $(\mathrm{I})_+ \sim (\mathrm{II})_-$
$(\mathrm{I})_- \sim (\mathrm{II})_+$

$\downarrow$



$\cong$ compact surface of genus 2.

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To compute the degrees of $z, w : X \longrightarrow \mathbb{P}^1$ we fix $p = \infty$.

$\deg(z) = z^{-1}(\infty)$. Using local coordinate near $z = \infty$

$$t = \begin{cases} \frac{1}{\sqrt{z}} & z \in (\mathrm{I}) \\ -\frac{1}{\sqrt{z}} & z \in (\mathrm{II}) \end{cases},$$

we have $z = \frac{1}{t^2}$ so $\deg(z) = 2$. Now

$$w = \begin{cases} \sqrt{z(z-1)(z-\lambda_1)(z-\lambda_2)(z-\lambda_3)} & z \in (\mathrm{I}) \\ -\sqrt{z(z-1)(z-\lambda_1)(z-\lambda_2)(z-\lambda_3)} & z \in (\mathrm{II}) \end{cases}.$$

$$= \frac{1}{t}\sqrt{\left(\frac{1}{t^2}-1\right)\left(\frac{1}{t^2}-\lambda_1\right)\left(\frac{1}{t^2}-\lambda_2\right)\left(\frac{1}{t^2}-\lambda_3\right)}$$

$$= \frac{1}{t^5}\sqrt{(1-t^2)(1-\lambda_1 t^2)(1-\lambda_2 t^2)(1-\lambda_3 t^2)}$$

which has a pole of order 5 at $t = 0$. So $\deg(w) = 5$.

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Abelian differentials of the third kind

X — cpct Riemann surface obtained by anal. ctn $\sqrt{z(z-1)(z-\lambda)}$

$Q_1, Q_2 \in X$. Then $\exists$ merom. $\omega_{Q_1Q_2}$ w simple poles at Q_1, Q_2

$$\operatorname{res}_{Q_1} \omega_{Q_1Q_2} = 1$$

$$\operatorname{res}_{Q_2} \omega_{Q_1Q_2} = -1.$$

Proof. if $Q_1, Q_2 \neq 0, 1, \lambda, \infty$ and $z(Q_1) \neq z(Q_2)$,

$$\omega_{Q_1Q_2} = \frac{cz+d+w}{(z-z(Q_1))(z-z(Q_2))} \frac{dz}{w} \qquad c, d \in \mathbb{C}.$$

$$\left.\begin{aligned} \text{st. } & cz(Q_1) + d + w(\tilde{Q}_1) = 0 \\ & cz(Q_2) + d + w(\tilde{Q}_2) = 0 \end{aligned}\right\} \text{ has soln. as } \begin{vmatrix} z(Q_1) & 1 \\ z(Q_2) & 1 \end{vmatrix} \neq 0.$$

In this case $\operatorname{res}_{Q_1} \omega_{Q_1Q_2} = \lim_{z \to z(Q_1)} \frac{cz+d+w}{z-z(Q_2)} \frac{dz}{w}$

$$= \frac{cz(Q_1)+d+w}{z(Q_1)-z(Q_2)} \frac{dz}{w}$$

if $z(Q_1) = z(Q_2)$, $\quad \omega_{Q_1Q_2} = \frac{1}{z-z(Q_1)} \frac{dz}{w}$

poles $z(Q_1), z(\tilde{Q}_1)$.

$$\omega'_{Q_1Q_2} = \frac{\frac{w_1-w_2}{z_2-z_1} z + \frac{z_1w_2 - z_2w_1}{z_2-z_1} + w}{(z-z_1)(z-z_2)} \frac{dz}{w}$$

$w_1 = w(\tilde{Q}_1) = 0 \Rightarrow$

$$\omega'_{Q_1Q_2} = \frac{\frac{w_2(z-z_1)}{z_1-z_2} + w}{(z-z_1)(z-z_2)} \frac{dz}{w}.$$

res at Q_2: $\lim_{z \to z_2} = \frac{\frac{w_2(z_2-z_1)}{z_1-z_2} + w(Q_2)}{(z_2-z_1)} \frac{dz}{w(Q_2)}$

$$= \frac{2}{z_2-z_1} dz.$$

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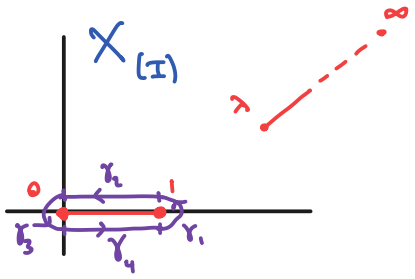
6. $\lambda \in \mathbb{C} \setminus \{0,1\}$. $\quad \Omega = \dfrac{dz}{\sqrt{z(z-1)(z-\lambda)}}$



parametrize γ as
$$\gamma_1 : [-\tfrac{\pi}{2}, \tfrac{\pi}{2}] \to \mathbb{C}, \quad t \mapsto 1+\epsilon e^{it}$$
$$\gamma_2 : [0,1] \to \mathbb{C}, \quad t \mapsto 1-t+i\epsilon$$
$$\gamma_3 : [\tfrac{\pi}{2}, \tfrac{3\pi}{2}] \to \mathbb{C}, \quad t \mapsto \epsilon e^{it}$$
$$\gamma_4 : [0,1] \to \mathbb{C}, \quad t \mapsto t-i\epsilon.$$

we compute
$$\left| \int_{\gamma_1} \frac{dz}{\sqrt{z(z-1)(z-\lambda)}} \right| = \left| \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{i\epsilon e^{it}}{\sqrt{(1+\epsilon e^{it})(\epsilon e^{it})(1-\lambda+\epsilon e^{it})}} dt \right|$$
$$\leq \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\epsilon}{|1+\epsilon e^{it}|^{\frac{1}{2}} \, \epsilon^{\frac{1}{2}} \, |1-\lambda+\epsilon e^{it}|^{\frac{1}{2}}} dt$$
$$\leq \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\epsilon^{\frac{1}{2}}}{(\frac{1}{2})^{\frac{1}{2}} \, (\frac{1}{2}|1-\lambda|)^{\frac{1}{2}}} dt, \quad \begin{array}{l} |1+\epsilon e^{it}| \geq 1-\epsilon > \frac{1}{2} \\ |1-\lambda+\epsilon e^{it}| \geq |1-\lambda|-\epsilon > \frac{1}{2}|1-\lambda| \end{array} \text{ for } \epsilon \ll 1.$$
$$= \frac{2\pi \epsilon^{\frac{1}{2}}}{|1-\lambda|^{\frac{1}{2}}} \xrightarrow{\epsilon \to 0} 0.$$

Similarly
$$\left| \int_{\gamma_3} \frac{dz}{\sqrt{z(z-1)(z-\lambda)}} \right| = \left| \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{i\epsilon e^{it}}{\sqrt{\epsilon e^{it}(\epsilon e^{it}-1)(\epsilon e^{it}-\lambda)}} dt \right|$$
$$\leq \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{\epsilon}{\epsilon^{\frac{1}{2}} \, |\epsilon e^{it}-1|^{\frac{1}{2}} \, |\epsilon e^{it}-\lambda|^{\frac{1}{2}}} dt$$
$$\leq \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{\epsilon^{\frac{1}{2}}}{(\frac{1}{2})^{\frac{1}{2}} \, (\frac{1}{2}|\lambda|)^{\frac{1}{2}}} dt, \quad \begin{array}{l} |\epsilon e^{it}-1| \geq |\epsilon-1| > \frac{1}{2} \\ |\epsilon e^{it}-\lambda| \geq |\epsilon-\lambda| > \frac{1}{2}|\lambda| \end{array} \text{ for } \epsilon \ll 1.$$
$$= \frac{2\pi \epsilon^{\frac{1}{2}}}{|\lambda|^{\frac{1}{2}}} \xrightarrow{\epsilon \to 0} 0.$$

Now
$$\int_{\gamma_2} \frac{dz}{\omega(z)} = \int_0^1 \frac{-dt}{\omega(1-t+i\epsilon)} = \int_0^1 \frac{-dt}{-\omega(1-t-i\epsilon)} \overset{(*)}{=} \int_1^0 \frac{-du}{\omega(u-i\epsilon)}$$
$$= \int_0^1 \frac{du}{\omega(u-i\epsilon)}$$
$$= \int_{\gamma_4} \frac{dz}{\omega(z)}.$$

(*) $u = 1-t$.

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Hence $\displaystyle\int_\gamma \frac{dz}{\omega(z)} = \sum_{i=1}^{4} \int_{\gamma_i} \frac{dz}{\omega(z)} \xrightarrow{\epsilon \to 0} -2\int_0^1 \frac{dt}{\sqrt{t(t-1)(t-\lambda)}}$

It's enough to show $\displaystyle\int_0^1 \frac{dt}{\sqrt{t(t-1)(t-\lambda)}} \neq 0.$

$$\text{this is} = \int_0^1 \frac{dt}{i\sqrt{t(1-t)(t-\lambda)}}$$

$$= -i\int_0^1 \frac{1}{\sqrt{t}\sqrt{1-t}}\, e^{-\frac{1}{2}\log(t-\lambda)}\, dt$$

$$= -i\int_0^1 \frac{1}{\sqrt{t}\sqrt{1-t}}\, e^{-\frac{1}{2}\log|t-\lambda|}\, e^{-\frac{1}{2}i\arg(t-\lambda)}\, dt$$

the argument of the integrand is $-\frac{1}{2}\arg(t-\lambda)$.

since $\lambda \neq 0, 1$, $t-\lambda$ is a line which does not pass through the origin, so it is contained in some half-plane of $\mathbb{C}$; say $\arg(t-\lambda) \in (\alpha, \alpha+\pi)$. Then

$$-\frac{1}{2}\arg(t-\lambda) \in \left(\beta, \beta + \frac{\pi}{2}\right) \quad \text{where } \beta = -\frac{\alpha}{2} - \frac{\pi}{2}.$$

Then

$$\arg\left(e^{-i(\beta+\frac{\pi}{4})} \frac{dt}{\sqrt{t(t-1)(t-\lambda)}}\right) \in \left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$$

So

$$\mathrm{Re}\left(e^{-i(\beta+\frac{\pi}{4})} \frac{dt}{\sqrt{t(t-1)(t-\lambda)}}\right) > 0,$$

and thus

$$\mathrm{Re}\left(e^{-i(\beta+\frac{\pi}{4})} \int_0^1 \frac{dt}{\sqrt{t(t-1)(t-\lambda)}}\right) = \int_0^1 \mathrm{Re}\left(e^{-i(\beta+\frac{\pi}{4})} \frac{dt}{\sqrt{t(t-1)(t-\lambda)}}\right) > 0$$

It follows that

$$\int_0^1 \frac{dt}{\sqrt{t(t-1)(t-\lambda)}} \neq 0, \quad \text{as desired.}$$

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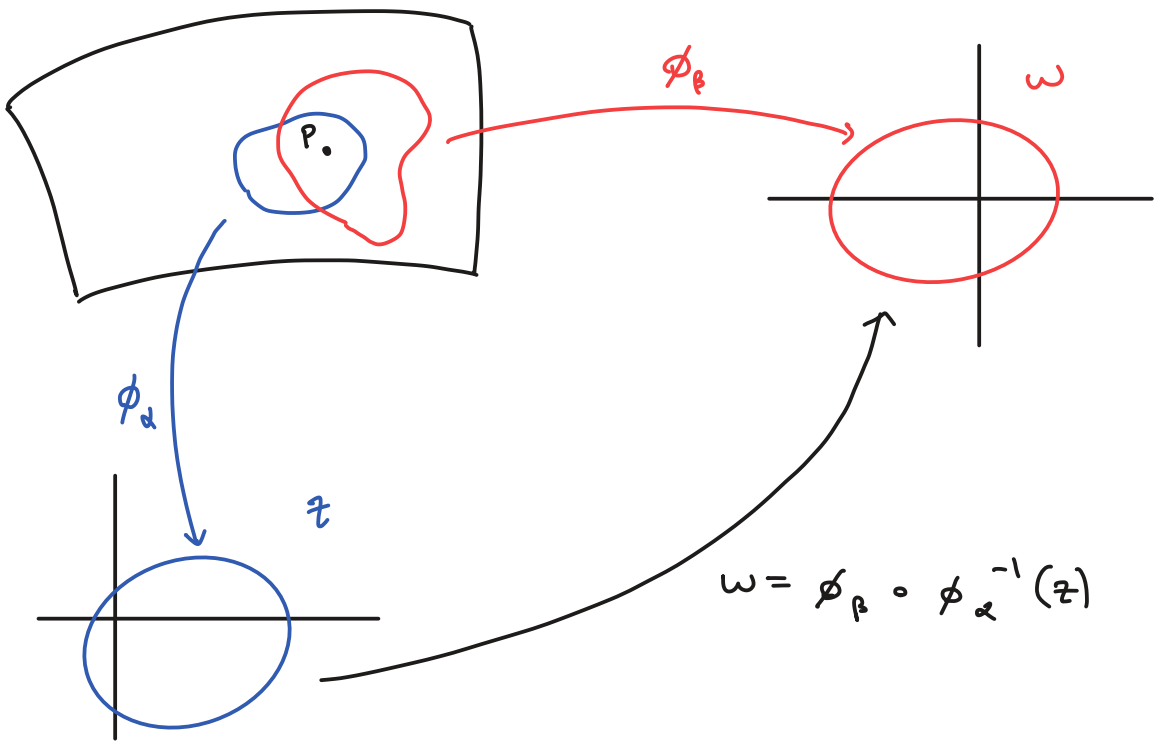

$$w = \phi_\beta \circ \phi_\alpha^{-1}(z)$$

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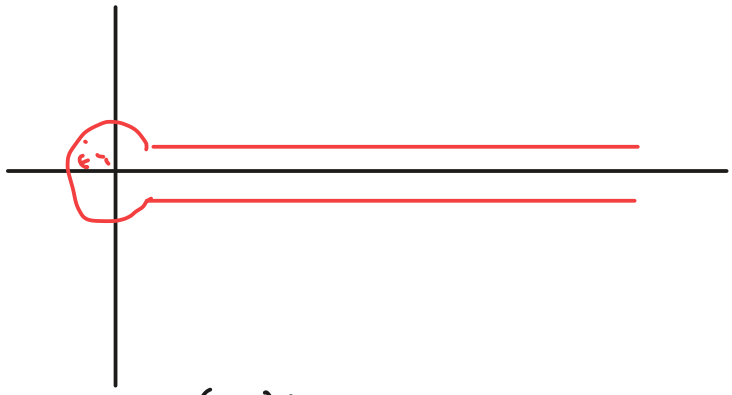
$\zeta(z)$ — analytic continuation of Riemann zeta fn. Show $\zeta(0) = -\frac{1}{2}$.

Proof. $s = \sigma + it$, $\sigma > 1$. For $\epsilon > 0$ let $\theta_\epsilon = \arcsin\left(\frac{\epsilon}{\sqrt{2}\epsilon}\right)$, and integrate $\frac{(-z)^{s-1}}{e^z - 1}$ along



Note $(z)^{s-1} = e^{(s-1)\log z}$ defined on $\mathbb{C} \setminus \mathbb{R}_{\leq 0}$ so the integral is parameterized by

$$\gamma_1 : [\epsilon, \infty) \to \mathbb{C}$$
$$t \mapsto t + i\epsilon$$

$$\gamma_2 : [\theta_\epsilon, 2\pi - \theta_\epsilon] \to \mathbb{C}$$
$$t \mapsto \sqrt{2}\,\epsilon\, e^{it}$$

$$\gamma_3 : [\epsilon, \infty) \longrightarrow \mathbb{C}$$
$$t \longmapsto t - i\epsilon.$$

along γ_1: $(-z)^{s-1} = e^{(s-1)\log(-t-i\epsilon)} = e^{(s-1)(\log|t+i\epsilon| + \overbrace{i\arg(t-i\epsilon)}^{\to -\pi})} \xrightarrow{\epsilon \to 0} t^{s-1} e^{-i\pi(s-1)}$.

So $$\int_{\gamma_1} \frac{(-z)^{s-1}}{e^z - 1}\, dz \longrightarrow \int_\infty^\epsilon \frac{t^{s-1} e^{-i\pi(s-1)}}{e^t - 1}\, dt$$
$$= -e^{-i\pi(s-1)} \int_0^\infty \frac{t^{s-1}}{e^t - 1}\, dt.$$

Similarly, $(-z)^{s-1} = e^{(s-1)(\log|t-i\epsilon| + i\arg(-t+i\epsilon))} \longrightarrow t^{s-1} e^{i\pi(s-1)}$ so

$$\int_{\gamma_2} \longrightarrow \int_0^\infty \frac{t^{s-1} e^{i\pi(s-1)}}{e^t - 1}\, dt$$

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$$\mathbb{P}^1 \longrightarrow \mathbb{P}^1$$

$$z \longmapsto \frac{az+b}{cz+d}, \qquad a,b,c,d \in \mathbb{C} \text{ ẽ } ad-bc=1$$

$$z_i, w_i \in \mathbb{P}^1 \rightsquigarrow f(z) = \frac{(z-z_1)(z_2-z_3)}{(z-z_3)(z_2-z_1)} \qquad \begin{cases} z_1 \longmapsto 0 \\ z_2 \longmapsto 1 \\ z_3 \longmapsto \infty \end{cases}$$

$$g(z) = \frac{(z-w_1)(w_2-w_3)}{(z-w_3)(w_2-w_1)} \qquad \begin{cases} w_1 \longmapsto 0 \\ w_2 \longmapsto 1 \\ w_3 \longmapsto \infty. \end{cases}$$

$$g^{-1}(w) = \frac{(w_2-w_1)w + w_1(w_2-w_3)}{}$$

$$(z - \infty) \rightsquigarrow \frac{1}{w} - \infty$$

$$cz_3 + d = 0 \Rightarrow d = -cz_3$$

$$bc = 1$$
$$\Downarrow$$
$$b = \frac{1}{c}$$

$$b = cz_2 + d$$

$$\frac{1}{c} = cz_2 + d \rightsquigarrow 1 = c^2 z_2 - c^2 z_3$$
$$= c^2(z_2 - z_3)$$
$$c^2 = \frac{1}{z_2 - z_3}$$

$$c =$$

$$a = c \neq 0$$

$$az_1 + b = 0 \rightsquigarrow b = -az_1$$
$$az_3 + d = 0 \rightsquigarrow d = -az_3. \quad \Big\} \; d - b = -az_3 + az_1$$
$$= a(z_1 - z_3) = \frac{1}{a}$$
$$\Rightarrow a^2 = \frac{1}{\sqrt{z_1 - z_3}}$$

$$a(d-b) = 1$$
$$d - b = \frac{1}{a}$$

$c = 0.$ $\quad \frac{az+b}{d}$ $\quad \frac{az_1}{d} + \frac{b}{d} = 0 \rightsquigarrow az_1 + b = 0 \rightsquigarrow b = -az_1$

$$a(z_2 - z_1) = az_2 + b = d = \frac{1}{a} \rightsquigarrow a = \frac{1}{\sqrt{z_2 - z_1}}.$$

$$ad = 1$$

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$$\operatorname{Im}\left(\frac{az+b}{cz+d}\right), \qquad a,b,c,d \in \mathbb{R}, \quad ad-bc=1$$

$$= \frac{ad-bc}{|cz+d|^2}\operatorname{Im}(z) > 0.$$

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